

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ParkNow – Smart Parking Aggregator with Real-Time Space Pre-Booking Matrix

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1. Introduction

Project Overview

ParkNow is a Smart Parking Management System that helps users find, reserve, and access parking spaces efficiently. The application provides real-time parking availability, online slot booking, secure QR-based entry, and centralized parking management through dedicated mobile applications for **Users**, **Staff**, and **Administrators**.

Built using **React Native** and **Supabase**, the system offers a cloud-based architecture with real-time synchronization, secure authentication, and scalable database management.

Purpose

The purpose of ParkNow is to reduce the time spent searching for parking spaces while improving parking space utilization through digital booking and real-time monitoring.

Scope

ParkNow serves three primary user roles:

Driver (User)

- Search nearby parking locations
- View available parking slots
- Reserve parking spaces
- Complete online payments
- Receive QR parking pass
- Navigate to parking location
- View booking history

Staff

- Register walk-in customers
- Assign parking slots
- Scan QR codes
- Verify vehicle entry
- Manage check-in and check-out
- Collect payments

Administrator

- Monitor parking occupancy
 - Manage parking locations
 - Manage parking slots
 - Manage staff accounts
 - View reports and statistics
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2. System Overview

ParkNow follows a **Client–Server Architecture**.

The mobile application communicates directly with **Supabase Backend**, which provides:

- PostgreSQL Database
- Authentication
- Real–Time Synchronization
- Row Level Security
- Cloud Storage

The system is divided into three independent workspaces:

- User Workspace
- Staff Workspace
- Administrator Workspace

Each workspace communicates through a centralized Service Layer to ensure modularity and scalability.

3. Functional Requirements

The major functionalities of ParkNow include:

Authentication

- Secure role–based login
- User registration
- Role verification

Parking Discovery

- Search nearby parking locations
- Interactive map
- Live parking availability

Slot Booking

- View live parking layout
- Select preferred parking slot
- Reserve parking space
- Automatic slot status update

Digital Payments

- Booking confirmation
- Parking fee calculation
- Digital payment processing
- Receipt generation

QR Verification

- Generate QR parking pass
- Scan QR at entry
- Vehicle verification
- Automatic status update

Staff Operations

- Walk-in booking
- Manual slot assignment
- Payment collection
- Vehicle check-in

Administration

- Dashboard
- Parking slot management
- Staff management
- Live occupancy monitoring

4. Non-Functional Requirements

Performance

- Fast response time
- Instant real-time updates

Security

- Secure authentication
- Role-Based Access Control
- Row Level Security

Reliability

- Accurate booking records
- Data consistency
- Secure transactions

Scalability

- Supports multiple parking facilities
- Cloud-based infrastructure
- Concurrent users

Usability

- Simple and intuitive interface
 - Responsive mobile design
 - Easy navigation
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5. Database Design

The system uses **PostgreSQL** with the following core tables:

- Roles
- Users
- Staff Profiles
- Parking Locations

- Parking Slots
- Vehicles
- Bookings
- Payments
- Verification Logs
- Staff Assignments

Parking Slot Status

Available



Reserved



Occupied



Available

Administrators can also mark slots as **Maintenance**.

6. Core Modules

User Module

The User Module allows customers to:

- Login
- Search parking
- View live slots
- Select parking slot
- Book parking
- Make payment
- Receive QR Pass

- Navigate to destination
 - View booking history
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Staff Module

The Staff Module enables staff members to:

- Register walk-in customers
 - Assign parking slots
 - Verify QR codes
 - Check-in vehicles
 - Collect payments
 - Monitor bookings
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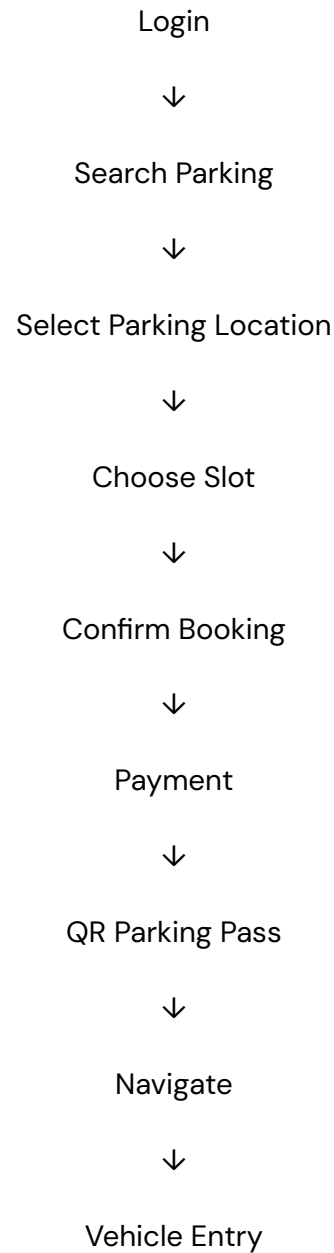
Administrator Module

The Administrator Module provides:

- Dashboard
 - Parking slot management
 - Staff management
 - Parking analytics
 - Occupancy monitoring
 - System administration
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7. System Workflow

User Booking Flow



Staff Workflow

Login



Manual Booking



Assign Slot



Collect Payment



Generate Booking



Vehicle Entry

QR Verification Flow

Scan QR Code



Validate Booking



Update Booking Status



Update Parking Slot



Allow Vehicle Entry

8. Real-Time Synchronization

ParkNow uses **Supabase Realtime** to synchronize parking information instantly.

Real-time updates include:

- Parking slot booking
- Slot availability
- Booking status
- Vehicle check-in
- Occupancy statistics
- Dashboard updates

This eliminates manual refreshes and ensures accurate parking information.

9. Technology Stack

Frontend

- React Native

Programming Languages

- JavaScript
- TypeScript

Backend

- Supabase

Database

- PostgreSQL

Authentication

- Supabase Authentication

Real-Time

- Supabase Realtime

Maps

- React Native Maps
- Leaflet

QR Code

- react-native-qrcode-svg
 - react-native-camera-kit
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10. Future Scope

Future improvements include:

- AI-Based Parking Prediction
 - Automatic Number Plate Recognition (ANPR)
 - IoT Parking Sensors
 - Electric Vehicle Charging Support
 - Dynamic Pricing
 - Google Maps Navigation
 - Multi-City Deployment
 - Advanced Analytics Dashboard
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11. Conclusion

ParkNow is a modern **Smart Parking Management System** that combines **React Native**, **Supabase**, and **PostgreSQL** to deliver an efficient, secure, and scalable parking solution. The application provides real-time parking availability, digital booking, QR-based vehicle verification, and role-based management for users, staff, and administrators.

With its cloud-based architecture, real-time synchronization, and modular design, ParkNow improves parking efficiency, reduces traffic congestion, and enhances the overall parking experience. The system is scalable and can be extended with Artificial Intelligence, IoT

integration, EV charging support, and smart city infrastructure, making it a practical solution for future urban parking management.